ELSEVIER

Earth and Planetary Science Letters 159 (1998) 133–145

EPSL

A palaeomagnetic study from the Mongol–Okhotsk region: rotated Early Cretaceous volcanics and remagnetized Mesozoic sediments

Nadir Halim [a,*], Vadim Kravchinsky [b], Stuart Gilder [a], Jean-Pascal Cogné [a], Maxim Alexyutin [c], Andrey Sorokin [d], Vincent Courtillot [a], Yan Chen [e]

[a] *Laboratoire de Paléomagnétisme, Institut de Physique du Globe de Paris, 4 place Jussieu, 75252 Paris Cedex 05, France*
[b] *East-Siberian Research Institute of Geology, Geophysics and Mineral Resources, and Institute of Geochemistry, Siberian Branch of the Russian Academy of Sciences, Irkutsk 664033, Russia*
[c] *Palaeomagnetic Laboratory, Geological Institute, Russian Academy of Sciences, Pyzhevsky Lane, 7, Moscow 109017, Russia*
[d] *Regional Geology and Hydrology Division, Amur Research Centre of the Russian Academy of Sciences, 675006 Blagovestzhenk, Russia*
[e] *Laboratoire de Géologie Structurale, Département des Sciences de la Terre, Université d'Orléans, B.P. 6759, 45067 Orléans Cedex 02, France*

Received 18 November 1997; accepted 30 March 1998

Abstract

We collected 47 sites of Upper Triassic to Middle Jurassic siltstones and sandstones and two sites of Lower Cretaceous andesites in a large basin located south of the Mongol–Okhotsk suture. The suture separates the Siberian craton to the north from the Mongolian and Chinese blocks to the south. Laboratory treatment and analyses identify the same post-folding direction in all rocks. The mean direction (with a corresponding palaeopole at 76.8ºN, 152.2ºE, $A_{95} = 4.2^{o}$) of the overprint component ($N = 49$ sites) is significantly different at the 95% confidence level from the expected time-averaged Brunhes or present-day field directions. The Lower Cretaceous andesites possess a stable remanent direction at high temperatures that is distinct at 95% confidence limits from the overprint direction. The corresponding pole (58.3ºN, 51.0ºE, dp/dm $= 3.8^{o}/4.6^{o}$), based on only fourteen samples, is significantly rotated $78.4^{o} \pm 5.3^{o}$ counterclockwise and insignificantly displaced $2.7^{o} \pm 3.4^{o}$ north with respect to the Early Cretaceous reference pole for Siberia. We argue that the rotation is likely tectonic in nature and not due to a chance reading of the palaeosecular variation of the Earth's magnetic field. Both the palaeomagnetic data and the folding patterns we observed in the field suggest that deformation associated with the suture continued after the Early Cretaceous and involved sinistral shear. © 1998 Elsevier Science B.V. All rights reserved.

Keywords: Okhotsk region; Mongolia; suture zones; andesites; Pangaea; pole positions; rotation

1. Introduction

The Eurasian tectonic mosaic is composed of numerous cratons and fold belts whose histories have been a topic of active palaeomagnetic research since the late 1950s [1] (Fig. 1a). However, despite over 30 years of investigation, the region north of 45ºN latitude, which comprises 60% of the Asian landmass, is practically devoid of reliable Mesozoic palaeomagnetic data. To help remedy this situation, we collected nearly 400 Upper Triassic to Lower Cretaceous samples, just south of the central segment of the Mongol–Okhotsk suture that separates the

* Corresponding author. Tel.: +33 14427-6870; Fax: +33 14427-7463; E-mail: halim@ipgp.jussieu.fr

0012-821X/98/$19.00 © 1998 Elsevier Science B.V. All rights reserved.
PII S0012-821X(98)00072-7

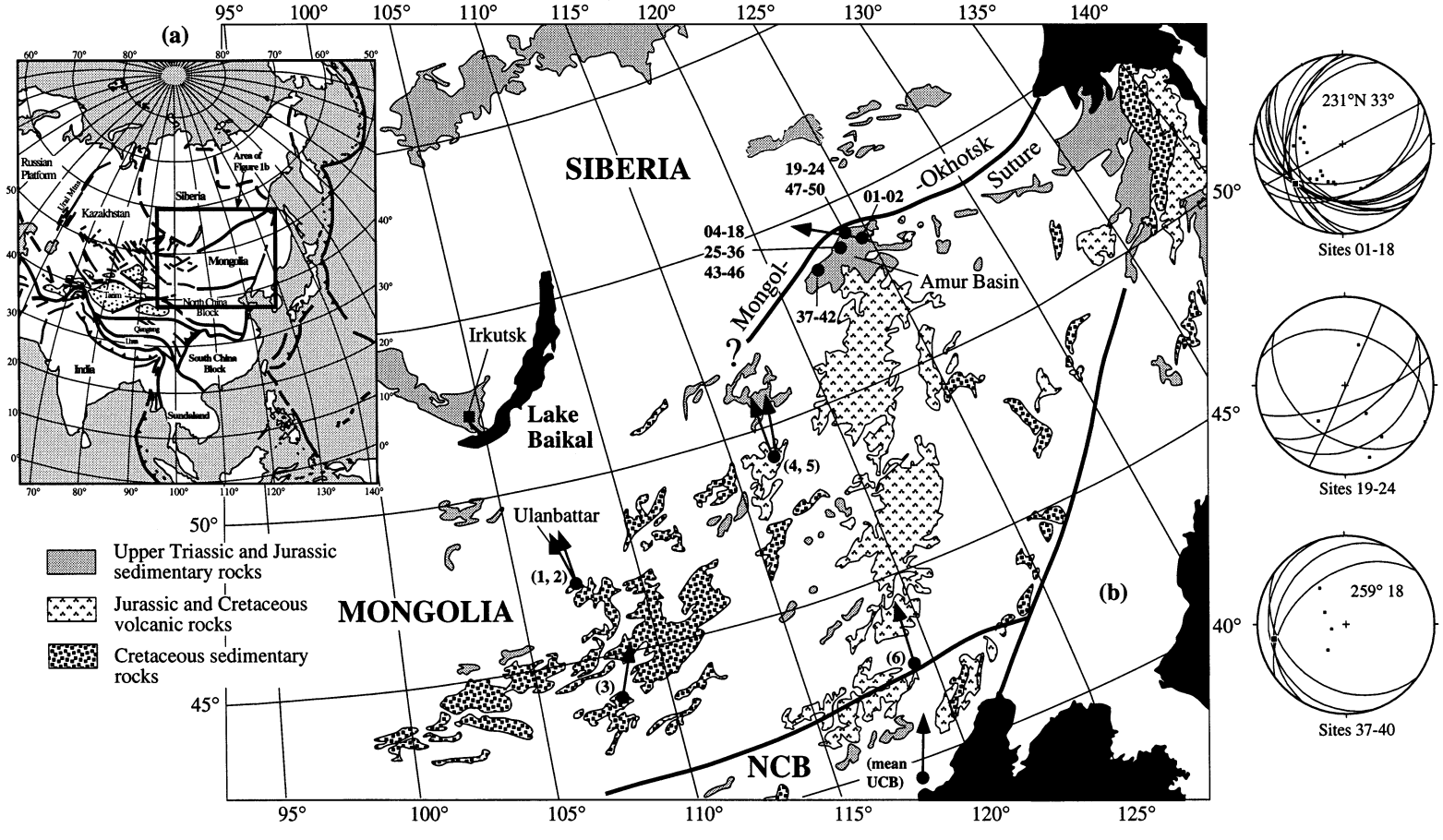


Fig. 1. (a) Simplified tectonic map of Asia. (b) Geological map of the Mongol–Okhotsk suture area. The region of Transbaikalia as referred to in the text extends from Lake Baikal to 122°E longitude and from 50 to 56°N latitude. Bold lines are tectonic boundaries. Arrows indicate the Cretaceous declinations from the Mongolian block. Numbers in parentheses correspond to Table 2. Numbers next to sampling sites have no parentheses. To the right are stereonet plots of the bedding orientations from the field areas of this study (Paris data only).

Siberian craton to the north from the Mongolian (Amurian) and Chinese blocks to the south. The Mongol–Okhotsk suture stretches along the southern boundary of the Siberian platform from the Sea of Okhotsk into central Mongolia (Fig. 1). The suture formed as a result of continental collision between Siberia and the lithospheric blocks to the south. The relative position of these blocks during the Palaeozoic and Mesozoic and their accretion or collision ages are subjects of active research (e.g. [2–6]) and in some cases the proposed reconstructions, including those related to the Mongol–Okhotsk suture, are hotly debated [7,8].

Misnik and Shevchuk [9] interpreted the rock assemblage near the town of Sretensk (118°E, 52°N) in the central part of the Mongol–Okhotsk suture as an ophiolite. Kuzmin and Fillipova [10] were among the first to publish on the history of the Mongol–Okhotsk suture in a plate tectonic framework. Subsequently, Parfenov [2] interpreted the geological data from the eastern segment of the suture consistent with a subduction model of oceanic crust between Siberia and the tectonic blocks to its southeast. Near the field area of this study ophiolites are described in the Skovorodino region (124.0°E, 53.8°N) [11,12]. They contain both mid-oceanic ridge (MORB) and oceanic island (OIB) basalts, as well as hyperbasic rocks, gabbro-tonalites, and plagiogranites [13]. Some of these complexes may be related to volcanic arc environments. Felsic intrusives parallel to the suture are thought to young towards the east, going from Late Carboniferous–Permian in central Mongolia, to Triassic–Early Jurassic in west Transbaikalia, Late Jurassic in east Transbaikalia, and Cretaceous in Amur Province, in the far-eastern part of Russia [13,14]. Palaeontological data also support the existence of a Palaeozoic to Middle–Late Jurassic ocean between Siberia and Mongolia [15–17]. Sediments bearing marine fossils young in the same direction as the granites (from west to east); there are no open ocean sediments younger than Permian in central Mongolia, Early Jurassic in the Transbaikal region, and Middle–Late Jurassic in Amur Province [15]. Field mapping suggests that some of the ophiolites overthrust lower Mesozoic rocks, and serpentinite mélanges contain fragments of Jurassic granites [4].

Although few Mesozoic palaeomagnetic data from Siberia and most of Kazakhstan pass modern standards of reliability, Besse and Courtillot [18] constructed a synthetic apparent polar wander path (APWP) for Eurasia by reconstructing Pangea via ocean floor magnetic anomalies. Once reconstituted, the poles from Africa, North America, India, and Europe were combined into a master APWP. This master APWP was transferred to each plate, then the plates (and corresponding APWPs) were restored to the present reference frame. A comparison of the combined APWP from the Chinese and Mongolian blocks with the Eurasian reference path suggests that they were separated by about 2000 km in the Late Jurassic, then sutured together in the Cretaceous [5,6].

2. Geologic overview

Palaeomagnetic samples were collected from 49 sites just north of the Amur River which forms the political border between Russia and the People's Republic of China (Fig. 1b). Our Franco–Russian team sampled Upper Triassic to Middle Jurassic black to grey sandstones and siltstones, rich in organic material, from three locations: 30 km south of Skovorodino (124.0°E, 53.8°N, sites 04–18, 25–36, 43–46), along the Trans-Siberian railroad between Takhtamygda and Skovorodino (123.6°E, 54.1°N, sites 19–24, 47–50), and 50 km southwest of Yerofey Pavlovich (121.9°E, 54.0°N, sites 37–42). These sediments were deposited in a large basin (called the Upper Amur depression in the Russian literature [15] and the Xinganling or Ermuerhu basin in the Chinese literature [19]). From bottom to top, the strata are separated into formations called: Ogon (Upper Triassic–Carnian to Norian), Kovali (Lower Jurassic–Pliensbachian–Toarcian), Skovorodino (Lower to Middle Jurassic), Oshurka (Middle Jurassic–Aalenian–Bajocian), Usmanko (Middle Jurassic–Bajocian–Bathonian), and Uskali (Middle Jurassic–Bathonian–Callovian). These formations are dated by abundant plant, pelecypod, and belemnite fossils as described in Refs. [15,20–22]. The average combined thickness of these formations exceeds 5900 m. In general, sediment size coarsens upward through the section; however, the entire sedimentary package is so uniform in appearance that we could not distinguish among

the various formations in the field. The Mesozoic lithologies suggest that sedimentation occurred in a relatively shallow sea. Some workers describe them as a molasse complex that disconformably overlies folded early Palaeozoic rocks [15,20,23].

We also sampled two sites (01–02) in two Lower Cretaceous andesite flows in a quarry near the town of Taldan (124.5ºE, 53.8ºN). The volcanic rocks at Taldan overlie both Jurassic sediments and an ancient metamorphic complex. The base of the Cretaceous pile has basaltic volcanic breccias and agglomerate tuffs that contain xenoliths of basement material. The agglomerate tuffs are covered mostly by porphyritic andesite flows that attain a thickness of 1000 m. The upper part of the suite possesses a layer of terrigenous sandstones and tuffaceous conglomerates. This bed is rich in Lower Cretaceous fresh-water pelecypods such as *Limnocyrena ovalis* (Ramm.), *L. wangehinensis* (Grab.), *L. altiformis* (Grab.), *Lacustrina lepida* Ch. Kol., and flora such as *Phyllotheca sibirica* (Heer), *Coniopteris* sp. ex gr. *C. nympharum* (Heer), *Cladophlebis* sp., thus its age is considered Lower Cretaceous [15,21]. Sedimentary horizons interbedded within the thick volcanic flows enabled us to obtain bedding orientations.

Fig. 1 contains equal-area stereonets of the bedding attitudes measured in the field. In the first area (sites 04–18), orientations strike west to southwest with dips of 30 to 50º. Bedding is relatively constant over distances of kilometres and it is possible that the sequence in this area represents a series of thrust packages that are stacked on southwest-dipping, bedding-parallel thrust faults. The intersection of the bedding planes defines a fold axis oriented N231º and dipping 33º. We note that the bedding planes of the Cretaceous andesites have attitudes similar to those from the first area, suggesting that folding of the Jurassic to Lower Cretaceous units occurred after the Early Cretaceous. This is consistent with post-Early Cretaceous folding found by Davis et al. [24] in the Yinshan belt, which they attributed to the collision of Siberia and Mongolia. In the second area (sites 19–24), closer to the suture, the beds are tightly folded with bedding orientations changing over distances of tens of metres and with no consistent fold axes. In the third area (sites 37–40), which is farther from the suture than the second, folding again occurs on longer wavelengths and the bedding attitudes define a plunging fold axis (18º towards N259º) similar to that observed in the first area.

3. Sampling, laboratory experiments and palaeomagnetic results

Samples were collected both as oriented hand samples and as standard 2.5-cm diameter cores using a portable drill with core orientation employing both magnetic and sun compass bearings. Sun compass corrections are particularly important because sampling was often carried out near railroads. At one site, our readings differ up to 30º from the International Geomagnetic Reference Field direction. For the core samples, stepwise thermal and alternating field demagnetization (using an average of 17 demagnetization steps per sample) and magnetic measurements were carried out in a shielded room at the Institut de Physique du Globe de Paris using JR-5 spinner and three-axis CTF cryogenic magnetometers. Magnetic susceptibility was measured after each temperature step with a Molspin susceptibility meter. Hand-oriented samples were cut into two to four cubes, each with a volume of 8 cm^3. These samples underwent stepwise thermal demagnetization at the palaeomagnetic laboratories in Irkutsk and Moscow using ovens housed in three concentric μ-metal shields. The residual field is 4–8 nT in the centre of the ovens. The samples were demagnetized using 25ºC steps up to 600ºC (Irkutsk), or using 22 temperature steps until 500ºC (Moscow). JR-4 spinner magnetometers were used to measure the remanent magnetizations. Magnetic susceptibility was measured with a Kappa-bridge KLY-2 (Irkutsk). In all cases, Zijderveld [25] diagrams were constructed for each sample, results were analyzed using principal component analysis [26], and site-mean directions were calculated using Fisher [27] statistics.

4. Upper Triassic to Middle Jurassic formations

In nearly every sample, stepwise thermal demagnetization defines a component that trends north and steeply down in in-situ coordinates. We refer to this direction as component one (C1). C1 often decays

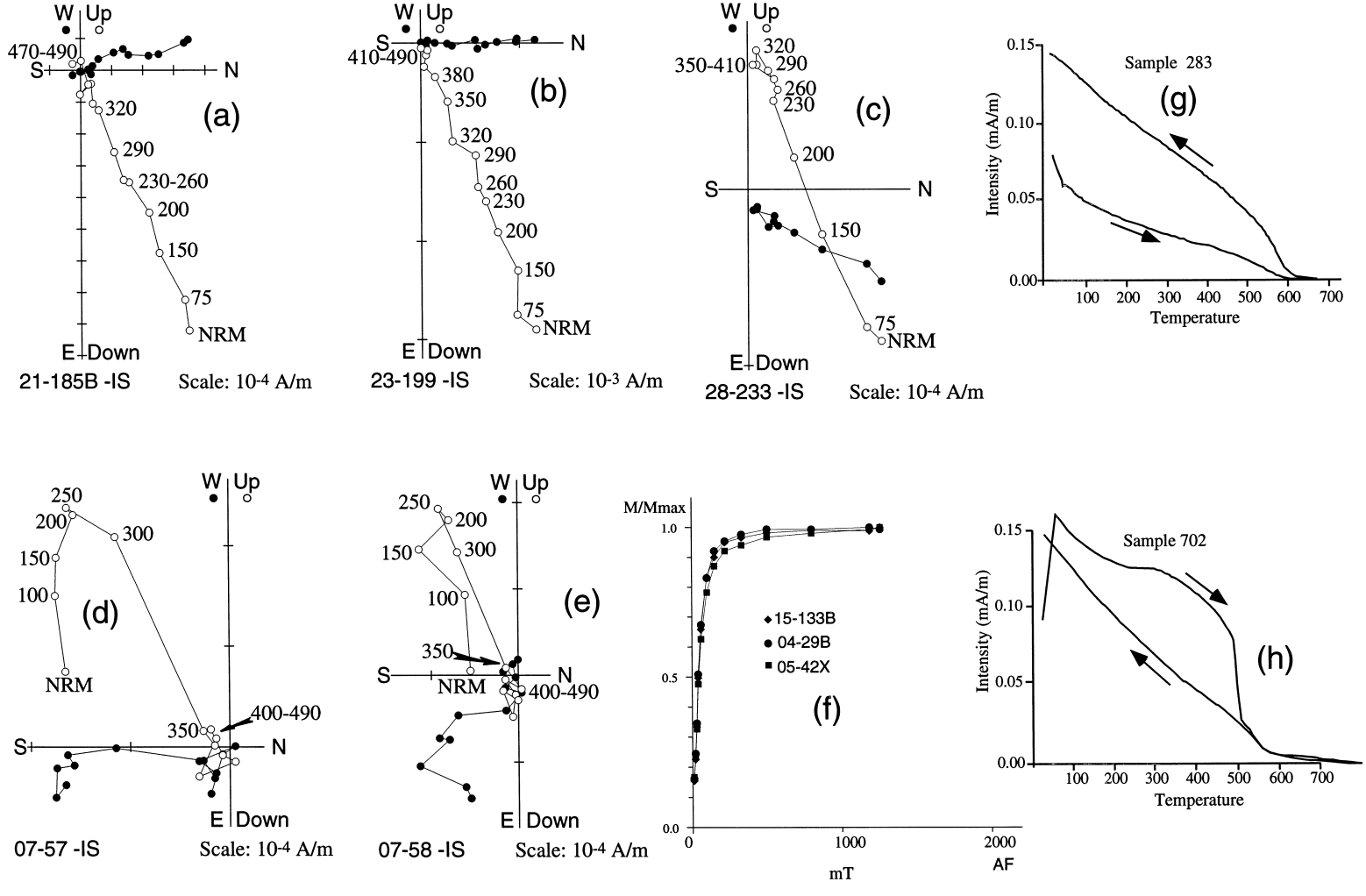


Fig. 2. (a–e) Orthogonal vector plots of C1-bearing samples — note reversed component in (d) and (e). (c) Orthogonal vector plot of a C1-bearing sample that does not decay toward the origin. (f) IRM acquisition data showing a rapid increase of magnetization until saturation is reached around fields of 300 mT. (g, h) Thermomagnetic curves of two sedimentary samples.

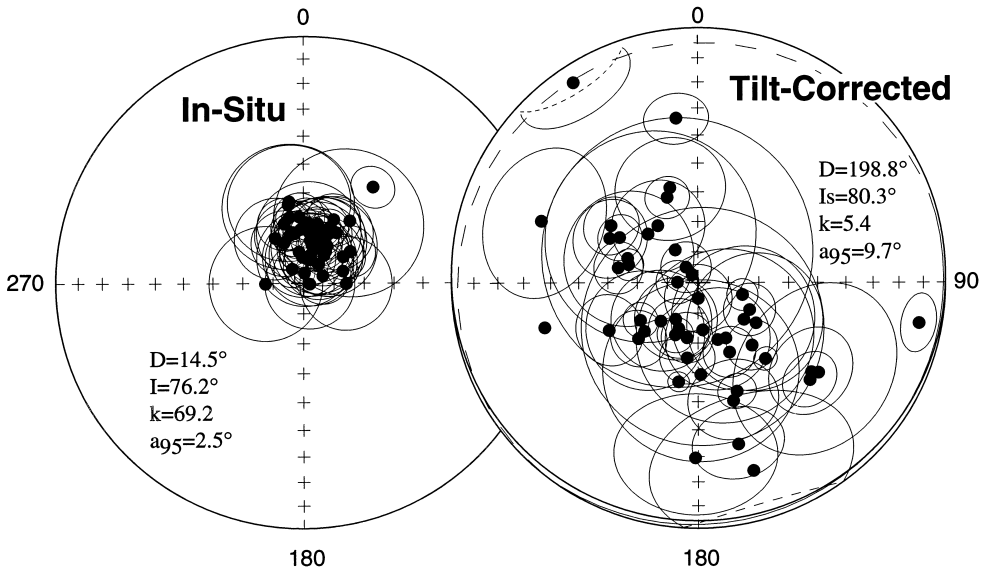


Fig. 3. Equal-area projection of the LTC + C1 site-mean directions with their 95% cones of confidence, before (left) and after (right) tilt correction.

toward the origin (Fig. 2a,b); sometimes it trends away from the origin during progressive unblocking until 440ºC, where after magnetization becomes unstable (Fig. 2c). The application of remagnetization great circles [28] to these samples was unsuccessful at defining anything but the C1 direction. Samples from sites 07 and 08 have two components of magnetization (Fig. 2d,e): one (C1) that trends north and steeply down (in-situ coordinates) at low temperatures (<250ºC) and a second that appears at higher temperatures. The higher temperature (>250ºC) direction is antipodal (reversed) to the C1 component in some cases (Fig. 2d,e), in other cases the directions are randomly oriented.

Complete unblocking occurs around 500ºC; however, susceptibility often increases above 400 to 450ºC and magnetization becomes unstable. NRM intensities range from 2×10^{-3} to 2×10^{-2} A/m except for site 22 whose intensity is unusually high (1 A/m). IRM experiments indicate that saturation of the remanence corresponds to bulk coercivities around 0.3 T (Fig. 2f). Curie point measurements show either a gradual decrease of magnetization from 25 to 580ºC (Fig. 2g) or a large decrease in the 500 to 580ºC temperature range (Fig. 2h). Taken together, these observations suggest that titanomagnetite is most likely the carrier of the remanence in these sediments.

For the site-mean C1 directions, the precision parameter (k) decreases from 67.6 in in-situ coordinates to 5.6 after tilt correction, clearly failing the fold test [29] (Fig. 3). Stepwise unfolding of the site-mean C1 directions ($N = 47$ sites) gives a maximum precision parameter (k) at 5% unfolding. Note that the fold test is negative also at the site level in several cases (Table 1).

5. Lower Cretaceous andesites

For the samples from both sites, stepwise thermal demagnetization isolates a low-temperature component (LTC) (<∼290ºC) which trends north and steeply down in in-situ coordinates, and a high-temperature component (HTC) (>∼410ºC), characterized by normal polarity directions that trend toward the origin upon progressive unblocking (Fig. 4a–d). The samples from site 01 contain an intermediate temperature component (ITC) (∼290 to ∼410ºC), characterized by directions with reversed polarity (Fig. 4a–c), that is nearly anti-parallel to the HTC (Table 1). Complete unblocking occurs at

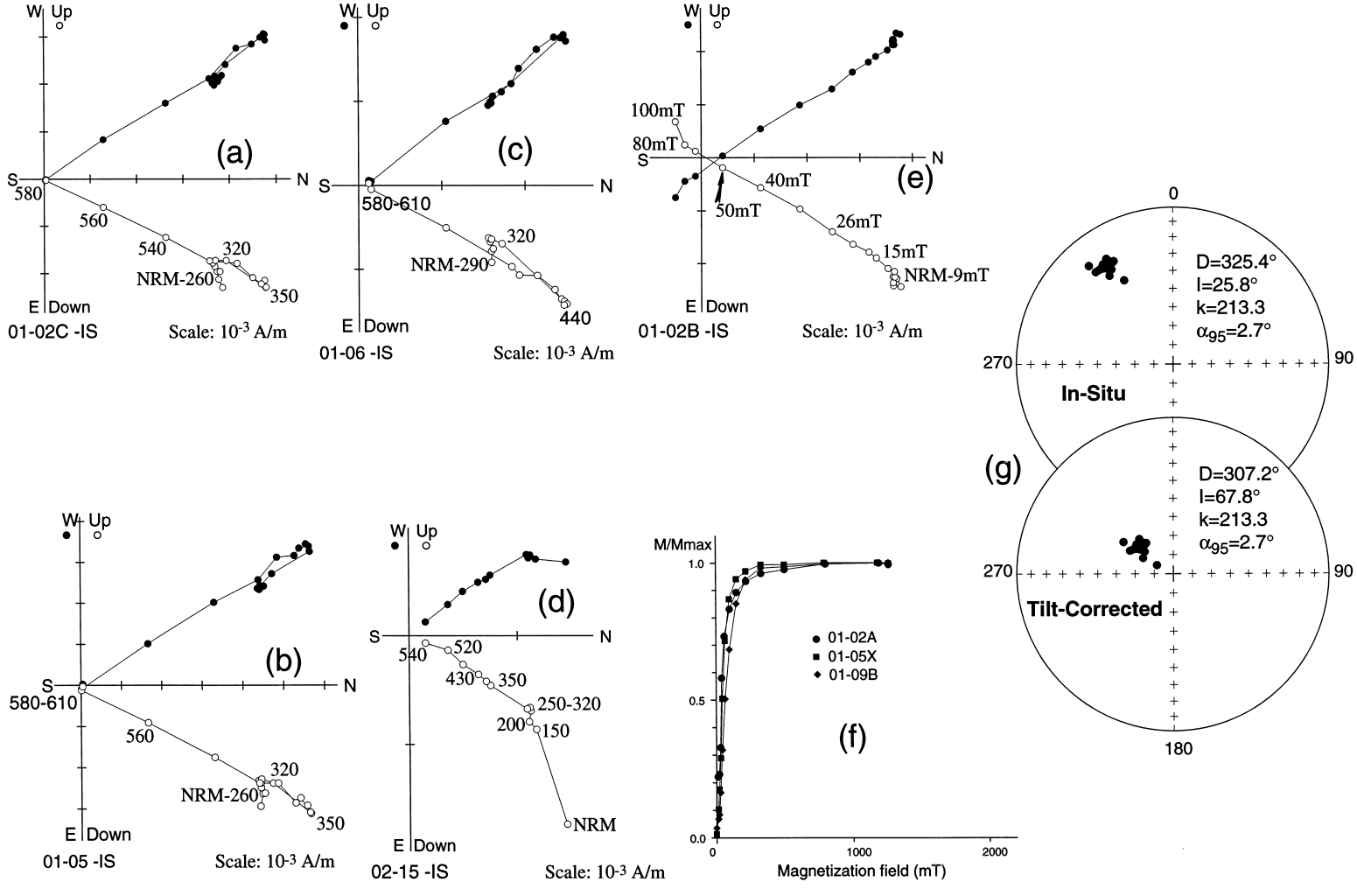


Fig. 4. (a–d) Thermal demagnetization orthogonal vector plots of four Cretaceous samples. (e) AF demagnetization vector plot of the sister specimen whose thermal demagnetization is shown in (a). (f) Results from IRM acquisition experiments of three Cretaceous samples. (g) Equal-area projection of the Cretaceous site-mean directions.

Table 1
Site-mean paleomagnetic directions for the magnetic components of Triassic, Jurassic and Cretaceous rocks

	Site	n	D_g	I_g	k	α_{95}	λ_p	ϕ_p	dp/dm	D_s	I_s	k	α_{95}
Cretaceous													
LTC	01-p	6	321.6	84.1	124.1	6.0	62.1	108.8	11.7/11.9	160.3	51.4	124.1	6.0
	02-p	5	19.6	74.1	393.3	3.9	77.6	174.7	6.3/7.0	138.6	56.2	393.3	3.9
ITC	01-p	6	141.6	−28.5	201.5	4.7	52.5	58.6	6.8/8.0	115.7	−68.6	201.5	4.7
HTC	01-p	7	326.6	25.2	396.2	3.0	60.1	48.8	4.2/5.1	310.5	67.6	396.2	3.0
	02-p	7	323.9	27.2	147.2	4.6	56.0	55.0	6.5/7.7	302.3	68.4	147.2	4.6
Mean HTC		14	325.4	25.8	213.3	2.7	58.3	51.0	3.8/4.5	307.2	67.8	213.3	2.7
Jurassic													
C1	04-p	5	5.7	86.7	502.2	3.4	60.3	125.1	6.8/6.8	191.1	56.3	502.2	3.4
	05-p	5	9.7	81.7	61.2	15.9	69.6	131.5	17.3/17.8	188.2	64.8	75.0	8.9
	06-p	6	56.0	71.5	64.4	8.4	58.7	186.4	12.9/14.7	191.3	71.4	64.4	8.4
	07-p	5	34.9	73.7	38.9	14.9	70.2	182.5	24.1/26.8	292.0	58.2	35.5	15.6
	08-p	4	36.1	50.3	170.1	7.1	55.6	240.5	6.4/9.5	271.1	83.3	170.1	7.1
	09-p	9	13.4	69.9	83.5	5.7	82.1	207.9	8.4/9.7	253.7	35.5	0.9	–
	10-p	6	356.8	72.4	49.6	9.6	85.8	99.7	15.1/17.0	227.7	66.0	12.6	19.7
	11-p	4	6.1	73.1	29.9	17.1	84.0	85.7	27.2/30.5	161.0	70.1	29.9	17.1
	12-p	8	20.8	71.8	87.4	6.0	77.9	191.7	9.2/10.5	203.4	71.2	87.4	6.0
	13-p	7	12.4	77.2	140.2	5.1	76.7	146.7	8.9/9.6	236.8	67.3	110.0	5.8
	14-p	6	15.6	73.8	58.8	8.8	79.6	172.5	14.3/15.9	226.7	62.9	53.0	9.3
	15-p	7	56.5	74.4	68.0	7.4	59.7	177.7	12.1/13.4	299.8	60.0	68.0	7.4
	16-p	5	73.0	76.5	229.3	5.1	53.3	167.8	8.7/9.4	284.5	66.3	229.3	5.1
	17-p	8	69.0	83.2	36.3	9.3	56.5	146.8	17.9/18.3	303.1	55.4	36.3	9.3
	18-p	7	29.1	76.6	53.5	8.3	71.9	166.1	14.3/15.4	280.5	63.1	53.5	8.3
	19-p	4	14.3	76.5	106.9	8.9	77.2	152.6	15.4/16.6	131.1	39.8	106.9	8.9
	20-p	4	349.7	73.5	151.6	7.5	82.2	81.4	12.1/13.4	100.4	10.2	151.6	7.5
	21-p	6	348.0	67.9	232.8	4.4	82.1	375.8	6.2/7.4	105.2	74.9	169.5	5.2
	22-p	3	356.4	74.2	91.1	13.0	83.2	108.6	21.3/23.5	11.0	35.1	91.1	13.0
	23-p	4	338.8	70.5	42.2	14.3	77.7	45.6	21.4/24.8	162.3	35.5	42.2	14.3
	24-p	6	344.4	72.5	82.2	7.4	80.5	63.4	11.6/13.1	243.8	56.9	82.2	7.4
	25-i	5	38.6	82.0	44.3	11.6	64.4	146.8	21.9/22.6	163.4	48.6	30.5	14.1
	26-i	9	23.5	76.0	52.2	7.2	74.5	165.7	12.2/13.3	155.7	64.7	38.0	8.5
	27-i	5	349.0	63.7	13.9	21.3	79.0	348.2	26.8/33.8	124.9	66.7	12.4	22.6
	28-i	7	13.8	73.6	26.1	12.0	80.5	171.2	19.4/21.6	179.1	84.8	30.4	11.1
	29-i	6	269.0	77.5	13.6	18.8	47.2	87.2	33.0/35.2	291.4	32.4	9.9	22.4
	30-i	4	350.0	62.6	22.4	19.8	78.2	341.5	24.2/31.0	153.5	69.5	10.9	29.2
	31-i	5	32.7	75.5	22.5	16.5	70.9	173.2	27.8/30.3	328.1	5.1	26.9	15.0
	32-i	4	334.0	75.4	73.5	10.8	73.9	76.8	18.1/19.8	202.7	73.6	86.5	9.9
	33-i	6	21.6	77.4	20.4	15.2	73.9	156.6	26.6/28.4	211.3	75.9	25.4	13.6
	34-i	6	341.8	69.5	59.8	8.7	79.2	38.1	12.7/14.9	117.8	70.8	61.3	8.6
	35-i	4	36.9	64.4	14.8	24.7	65.4	216.6	31.7/39.6	174.4	74.3	6.2	40.3
	36-i	7	33.3	70.2	25.8	12.1	70.6	198.9	18.0/20.9	128.8	70.6	7.4	23.8
	43-m	5	90.5	75.9	26.0	15.3	46.0	164.0	26.0/28.2	163.7	21.4	8.1	28.6
	44-m	5	97.1	87.9	27.8	14.8	53.1	130.7	29.5/29.5	180.9	29.4	11.7	23.4
	45-m	7	18.4	81.9	16.9	15.1	68.4	137.4	28.5/29.3	325.1	77.0	2.7	45.6
	46-m	8	24.2	74.9	15.3	14.6	75.0	172.6	24.3/26.6	289.0	65.4	7.3	21.9
	47-m	7	9.1	75.8	44.2	9.2	79.5	146.9	15.6/16.9	128.3	41.2	19.6	14.0
	48-m	7	352.0	79.7	86.6	6.5	73.4	114.2	11.9/12.4	178.3	59.4	4.6	31.5
	49-m	6	357.8	81.5	27.1	13.1	70.4	121.9	24.6/25.4	324.0	86.8	1.5	90.9
	50-m	6	36.6	78.7	27.9	12.9	67.7	159.5	23.1/24.4	126.8	39.1	6.9	27.6
Triassic													
	37-p	4	14.4	80.4	46.2	13.7	71.4	137.7	25.2/26.3	223.9	72.4	31.5	16.6
	38-p	5	35.4	77.4	92.5	8	68.9	164.2	14.0/15.0	322.1	83.7	85.9	8.3

Table 1 (continued)

	Site	n	D_g	I_g	k	α_{95}	λ_p	ϕ_p	dp/dm	D_s	I_s	k	α_{95}
	39-p	4	25.7	66.8	111.6	8.7	73.6	217.9	11.9/14.4	352.5	33.3	95.7	9.4
	40-p	5	25.5	69.7	161.1	6.0	75.0	203.8	8.9/10.4	340.2	60.1	115.5	7.1
	41-i	4	3.7	81.7	34.2	15.9	70.0	126.8	29.9/30.9	343.6	57.2	28.3	17.6
	42-i	7	356.5	68.2	55.4	8.2	86.8	346.1	11.6/13.8	324.7	67.1	16.5	15.3
Mean C1		47S	14.5	76.2	67.6	2.5	77.0	152.5	4.4 [a]	310.3	85.0	9.1	7.3
Mean LTC + C1		49S	14.7	76.3	69.2	2.5	76.8	152.2	4.2 [a]	198.8	80.3	5.4	9.7

Abbreviations are as follows: -p, -i, and -m, data from Paris, Irkutsk, or Moscow laboratories, respecively; n, number of treated specimens or sites (S); D_g, I_g (D_s, I_s), mean directions in geographic/in-situ (stratigraphic/tilt-corrected) coordinates; k and α_{95}, parameters of Fisher [27] statistics; λ_p (ϕ_p), latitude (longitude) of the virtual geomagnetic pole (VGP) corresponding to the direction observed at each site; dp/dm, minor/major axes of the 95% confidence ellipse. For the ITC and HTC (sites 01 and 02), VGPs are given in stratigraphic coordinates. For the LTC and C1, VGPs are given in in-situ coordinates.

[a] The radius of circle of significance that the mean pole lies in 95% confidence (A_{95}).

580ºC for most samples; some from site 02 have higher unblocking temperatures and probably contain hematite.

Alternating field (af) demagnetization of the ITC-bearing samples defines a northwest- and downward-directed component that is offset from the origin (Fig. 4e). Best-fit line segments of this component define a direction similar to that of the HTC, suggesting that the magnetic carrier of the HTC has a low coercivity. The natural remanent magnetization (NRM) direction after af demagnetization in fields with maximum amplitudes of 0.1 T is southeast and up, similar to the direction defined by the ITC. Total unblocking of the sample is not achieved by fields of 0.1 T. In fact, the magnetization intensity systematically increases after the sample has been subjected to applied fields of 60 mT. Acquisition of isothermal remanent magnetization (IRM) experiments show that the remanence saturates around applied fields of 0.3 T (Fig. 4f). Because the unblocking temperature of the ITC is higher (>400ºC) than expected for iron sulphides (pyrrhotite) and lower than expected for hematite, and because the bulk coercive force is low compared to most iron sulphides and hematite [30], it is doubtful that these minerals contribute to the magnetic signal.

Because the ITC is anti-parallel to the HTC, the ITC could be a self-reversed thermal remanent magnetization (TRM). Self reversing behaviour is found in some volcanic rocks of dacitic and andesitic compositions [31–33]. Such rocks can contain two ferromagnetic minerals, one of which is in the ilmenite–hematite series ($Fe_{2-y}Ti_yO_3$) with $0.45 \leq y \leq 0.65$. Curie temperatures corresponding to these compositions lie between 280 and 50ºC, which are the unblocking temperatures of most self-reversed TRMs found in nature. Here, the ITC persists to at least 400ºC. Two alternative explanations for the ITC are: (1) the rock contains two ferromagnetic minerals that grew at different times (e.g., primary crystallization followed by hydrothermal alteration), one during a period of normal polarity and the other when the Earth's field was reversed; or (2) site 01 was heated during a period of reversed polarity and grains with high bulk coercivities were preferentially thermally remagnetized during the heating event.

6. Discussion and conclusions

The main conclusion evinced from this palaeomagnetic study is that the Upper Triassic to Middle Jurassic sediments are overprinted. Because the pole to bedding of the Lower Cretaceous andesites lies in the same swath as the older sediments (Fig. 1), we assume that a single deformation event is responsible for the folding of all the rocks; thus the age of folding (and C1 remanence acquisition) is probably post-Early Cretaceous. The andesite LTC site mean direction is indistinguishable at 95% significance from the grouping of C1 site mean directions of the older sediments (Table 1) [34], in agreement with the suggestion that the remagnetization is a post-Early Cretaceous phenomenon. The combined LTC and C1 in-situ direction is $D = 14.7^o$, $I = 76.3^o$ ($\alpha_{95} = 2.5^o$, $N = 49$), with a correspond-

ing pole position at 76.8ºN, 152.2ºE (A_{95} = 4.2º) (Table 1).

The overprint pole is plotted with the Eurasian reference APWP in Fig. 5a. Because the age of overprinting is likely post-Early Cretaceous, the pole does not represent a rotated Late Jurassic overprint pole. With respect to the reference APWP, the overprint pole is indistinguishable within 95% confidence limits from the 20, 30, 40, and 50 Ma poles and is significantly different at 95% confidence from the 10 Ma and all pre-50 Ma poles [34]. However, except for four samples (~1%), the C1 component is dominated by directions with normal polarities, thus interpretation of the overprint as pre-Brunhes should be considered as tentative.

Although no field test exists for the andesites, we make the following observations.

(1) The expected Early Cretaceous inclination from the synthetic Eurasian APWP [18] is 69.5 ± 1.3º; the HTC tilt-corrected inclination (67.8 ± 2.7º) is indistinguishable from it while the in-situ inclination (25.8 ± 2.7º) is inconsistent with any post-Jurassic inclination predicted by the reference curve.

(2) Because the LTC of the andesites is indistinguishable from the C1 of the dark sediments, the LTC and C1 probably post-date the HTC.

(3) Because only site 01, which is stratigraphically below site 02, has an ITC, and because both site HTC directions are similar (Table 1), and if the ITC is not a self-reversed TRM, then the time elapsed between HTC acquisition of sites 01 and 02 spanned a period of more than two reversals.

Our Early Cretaceous andesite pole is rotated 78.4 ± 5.3º counterclockwise with respect to the mean Cretaceous reference pole from Eurasia (74.9ºN, 205.6ºE, A_{95} = 1.9º) (Fig. 5b and Table 2). One could question whether this relative rotation is an artifact of insufficient averaging of secular vari-

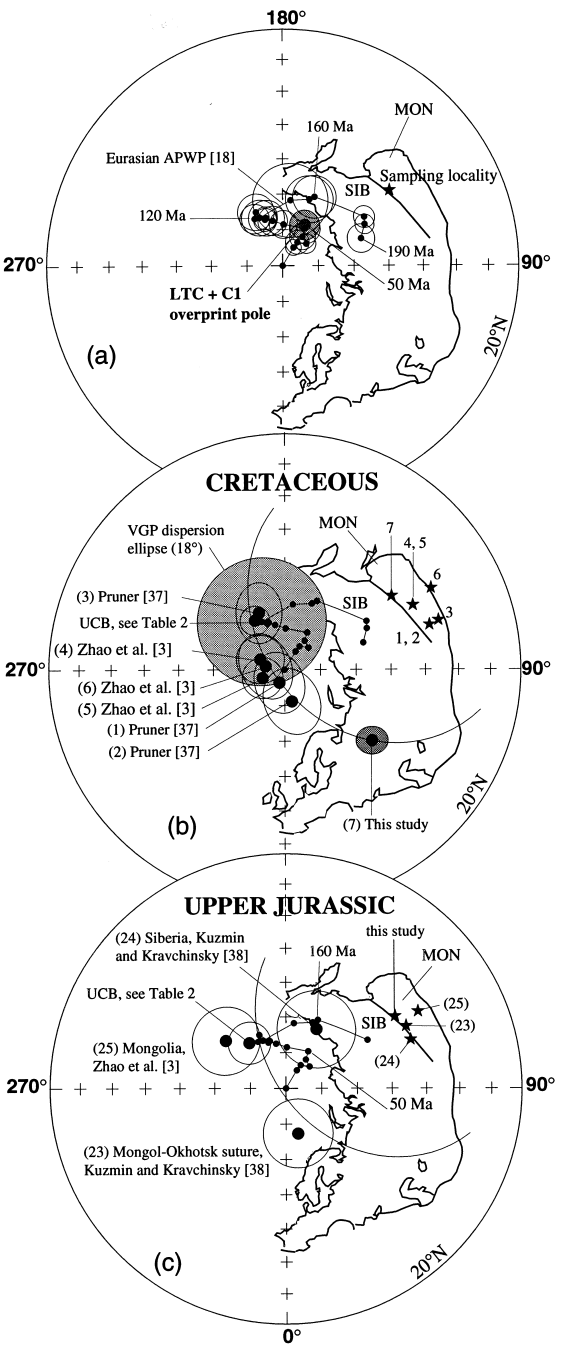


Fig. 5. (a) Equal-area projection of the mean LTC + C1 pole with the Eurasian APWP [18]. (b) Equal-area projection of the Cretaceous HTC pole together with other available Cretaceous poles from Mongolia, the mean Cretaceous pole from the united China block (*UCB*), and the Eurasian APWP. An error ellipse with a radius of 18º is drawn about the Cretaceous reference pole to represent palaeosecular variation. (c) Equal-area projection of the Eurasian APWP and the Late Jurassic poles from Mongolia, the UCB and Siberia. The Cretaceous and Jurassic poles are listed in Table 2. Numbers in parentheses in (b) and (c) correspond to Table 2; stars indicate the sampling localities corresponding to each pole.

Table 2
List of late Mesozoic palaeomagnetic poles from selected Asian tectonic blocks

Pole number	Pole			*N*	Block	Age	References
	Lat.	Long.	A_{95} (dp/dm)				
1	86.2	337.5	6.9/8.8	17s	MON	K	#12, [37]
2	80.9	12.7	7.7/9.7	19s	MON	K	#13, [37]
3	72.5	204.5	6.4/8.6	30s	MON	K	#14, [37]
4	82.6	247.3	6.7/8.6	11s	MON	K1	J1, [3][a]
5	83.4	290.3	5.7/7.4	9s	MON	K1	J2, [3][a]
6	84.6	259.8	7.0/9.6	4L	MON	K1	A7–A10, [3][a]
7	58.3	51.0	3.8/4.6	14s	M–O	K1	This study
8	74.9	205.6	1.9	4[b]	SIB	K1	[18]
9	81.5	220.9	7.1	18L	SCB	K3	[39]
10	66.0	221.5	3.4	43s	SCB	K3	[40]
11	56.2	211.5	3.9	19s	SCB	K3	[41]
12	66.9	221.4	5.4	5L	SCB	K3	[42]
13	72.8	241.1	6.6	16L	SCB	K3	[43]
14	77.1	227.6	5.5	7L	SCB	K1	[44]
15	74.5	229.0	4.0	23L	SCB	K1	[43]
16	77.4	196.2	14.5	11s	SCB	K1	[45]
17	69.0	204.6	4.3	7L	SCB	K1	[39]
18	78.2	171.9	10.6	12L	SCB	K1	[46]
19	79.6	170.1	5.8	4L	NCB	K1	[47]
20	74.5	203.4	8.2	10L	NCB	K1	[48]
21	75.8	208.7	7.5	10L	NCB	K1	[49]
22	74.5	201.0	4.7	10L	NCB	K1	[6]
mean (9–22)	73.9	211.5	4.1	14S	UCB	K	
23	77.0	15.0	7.8/9.7	26s	M–O	J3	[41]
24	71.0	153.0	9.9/10.8	26s	SIB	J3	[41]
25	68.5	231.6	9.5	4L	MON	J3	[3]
26	74.9	214.3	6.5	4S	UCB	J3	[6]

Abbreviations are as follows: Lat., latitude of locality; Long., longitude of locality; A_{95} (dp/dm), radius (minor and major axes) of the circle (ellipse) of confidence; *N*, number of samples (s), localities (L), or studies (S) used in the computation of the virtual geomagnetic pole; SIB, Siberia; MON, Mongolia; M–O, Mongol–Okhotsk suture; NCB, North China Block; SCB, South China Block; UCB, United China Block (NCB + SCB); [a] poles after Zhao et al. [3] calculated from tilt-corrected (not in-situ) directions; [b] the K1 pole for Siberia is computed using four poles of Early Cretaceous age (100, 110, 120 and 130 Ma) [18]; K, Cretaceous; J, Jurassic; 1, Early; 3, Late.

ation of the Earth's field (e.g., that the rotation is due to a chance reading of an instantaneous Early Cretaceous field direction). First, we note that a decimetre-thick sedimentary layer separates the two flows, so some time elapsed between eruptions. Second, we plotted the secular variation error ellipse (18º) calculated for a palaeolatitude of 50º [35] about the Cretaceous Eurasian reference pole in Fig. 5b. We note that 18º is a conservative estimate because the Early Cretaceous has been proposed as a time when secular variation was lower than today [36]. The andesite flows from this study are distinct at greater than 95% confidence limits from the secular variation ellipse. Because there are two flows, we consider that the probability that these directions are the result of a spot reading of a wobbling magnetic pole is very low.

Besides our data, there are a few other late Mesozoic palaeomagnetic data from the Mongolian plate [3,37,38] (Table 2). Like the pole from this study, the Early Cretaceous poles from Pruner [37] and Zhao et al. [3] were also derived from a limited number of samples/sites. The Late Jurassic poles from Zhao et al. [3] demonstrate positive fold tests and were calculated from site-mean data. The mean Late Jurassic poles from Kuzmin and Kravchinsky [38] were calculated at the sample level; although fold tests are not demonstrated, multi-component and dual polar-

ity magnetizations argue that the remanences are primary.

Because large rotations have only been found close to the suture zone (Figs. 1 and 5b,c), it is likely that the counter-clockwise rotation of sites 01 and 02 is local in nature and related to shear associated with the Mongolia–Siberia collision. The fact that the Late Jurassic pole corresponding to the locality north of the suture (Siberian craton, location 24, Fig. 5c) is concordant (not rotated) with respect to the coeval poles from the Eurasian reference APWP could be interpreted to suggest that the edge of the ancient Siberian craton was less deformed than the relatively younger Mongolian block to its south. In sum, the pattern of folding we observed in the field, together with the pattern of palaeomagnetic rotations, suggest that deformation related to collision continued after the Early Cretaceous and included a component of sinistral shear.

Acknowledgements

The field excursion and some laboratory work for this project were generously funded by INTAS, the Russian Foundation of Fundamental Research (96-05-64818), and the State Geological Company of Amur Province. The manuscript benefitted from computer programs coded by Jean-Pascal Cogné, Randy Enkin and Vinarsky and others. Konstantin Vakhtomin provided valuable guidance in the Siberian outback. NH and SG thank Vladimir, Ludmila Fiodarovna, and the Alexyutin family and SG thanks Natasha Levashova for their gracious hospitality. VK thanks M. Funaki for use of equipment. Mikhail Bazhenov, V.I. Golik, M.I. Kuzmin, and V.A. Rybalko aided various aspects of the work in Russia. Helpful reviews from Bill Harbert and Xixi Zhao are appreciated. This is Institut de Physique du Globe de Paris contribution 1526. ***[AC]***

References

[1] A.N. Khramov, Paleomagnetic Correlation of Sediment Formations, Gostechizdat, Leningrad, 1958, 218 pp.

[2] L.M. Parfenov, Continental Borders and Island Arcs of Mesozoic of the North-East Asia, Nauka, Novosibirsk, 1984, 189 pp.

[3] X. Zhao, R.S. Coe, Y. Zhou, H. Wu, J. Wang, New paleomagnetic results from northern China: collision and suturing with Siberia and Kazakhstan, Tectonophysics 181 (1990) 43–81.

[4] L.P. Zonenshain, M.I. Kuzmin, L.M. Natapov, Geology of the USSR: A Plate-Tectonic Synthesis, AGU, Geodyn. Ser. 21 (1990) 242 pp.

[5] R. Enkin, Z.Y. Yang, Y. Chen, V. Courtillot, Paleomagnetic constraints on the geodynamic history of China from the Permian to the Present, J. Geophys. Res. 97 (1992) 13953–13989.

[6] S. Gilder, V. Courtillot, A high resolution middle to late Mesozoic apparent polar wander path from the North China Block: implications for the North–South China collision, J. Geophys. Res. 102 (1997) 17713–17727.

[7] V. Courtillot, R. Enkin, Z. Yang, Y. Chen, M. Bazhenov, J. Besse, J.P. Cogné, R. Coe, X. Zhao, S. Gilder, Reply to Comment of S. Nie, D.B. Rowley, J. Geophys. Res. 99 (1994) 18043–18048.

[8] S. Nie, D.B. Rowley, Comment on 'Paleomagnetic constraints on the geodynamic history of the major blocks of China from the Permian to the Present' by R.J. Enkin et al., J. Geophys. Res. 99 (1994) 18035–18042.

[9] Yu.V. Misnik, V.V. Shevchuk, Eastern Trans-Baikal ancient block and its role for forming of regional structure, Geotectonica 5 (1980) 25–37.

[10] M.I. Kuzmin, I.B. Filippova, History of Mongol–Okhotsk folded belt development in Middle–Late Palaeozoic and Mesozoic, in: Structure of Lithosphere Plates, Inst. Oceanology Publ. House, Moscow, 1979, pp. 189–226.

[11] A.A. Sorokin, Geochemistry and Geodynamic Position of Magmatic Rocks of the Central Segment of the Mongol–Okhotsk Fold Belt, Ph.D. Thesis, Inst. Geochemistry, Siberian Branch of the Russian Academy of Sciences, Irkutsk, 1992, 156 pp.

[12] G.S. Gusev, A.I. Peskov, New data about Rifean ophiolites of Prishilkin segment of the Mongol–Okhotsk belt, Dokl. RAN 333 (1993) 220–224.

[13] A.A. Sorokin, S.I. Drill, M.I. Kuzmin, Geochemistry of rocks and paleogeodynamic position of the Yankan ophiolite complex of the Mongol–Okhotsk fold belt, in: Geodynamics and Evolution of the Earth, Siberian Branch of the Russian Academy of Sciences Publishing House, Novosibirsk, 1996, pp. 67–69.

[14] T.V. Molchanova, Role of Mesozoic granitoid magmatism in structures of Pacific ocean tectonic belt, Geotectonica 1 (1973) 18–97.

[15] Anonymous, Khabarovsk and Amur Provinces, in: L.I. Krasny (Ed.), Geology of the USSR, Nedra, Moscow, 1966.

[16] S.Z. Guo, B.H. Huang, K. Ichikawa, H. Inokuchi, K. Ishii, R.X. Liang, X.Y. Liu, T. Matsuda, J. Tazawa, Q. Wang, A. Yao, Pre-Jurassic evolution of Inner Mongolia, in: N.L. Dobretsov (Ed.), Report Nr. 6 of the IGCP Project 224 and Report Nr. 1 of the IGCP Project 283, Inst. Geol. Geophys. Publ. House, Siberian Branch of the Academy of Sciences of the USSR, Novosibirsk, 1990, pp. 34–35.

[17] V.G. Belichenko, E.V. Sklyarov, N.L. Dobretsov, O. Tomurtogoo, Geodynamic map of paleo-Asian ocean, east segment, Russ. J. Geol. Geophys. 7–8 (1994) 29–40.
[18] J. Besse, V. Courtillot, Revised and synthetic apparent polar wander path of the African, Eurasian, North American and Indian plates, and true polar wander since 200 Ma, J. Geophys. Res. 96 (1991) 4029–4050.
[19] Heilongjiang Bureau of Geology and Mineral Resources, Regional Geology of Heilongjiang Province, Geological Publishing House, Beijing, 1993, 734 pp.
[20] E.A. Kozlovsky (Ed.), Geology of the Baikal–Amur Rail Road Zone, VSEGEI, Nedra Publ. House, Leningrad, 1988, 443 pp.
[21] Conclusions of IV Inter-Organizational Regional Stratigraphic Meeting for Precambrian and Phanerozoic Epochs at South of Far East and Eastern Transbaikalia, Khabarovsk State Mine–Geological Co. Publ. House, Khabarovsk, 1990, 86 pp.
[22] R.T. Khomenko, Biostratigraphical study of Jurassic sea sediments of the West Amur province for geological survey, Report on theme 004 for 1990–1994 years, Amur Committee on Geol. and Min. Res. 'Amurgeology', Zeia, 1994, 86 pp.
[23] M.S. Nagibina, Stratigraphy and Formations of the Mongol–Okhotsk Belt, Geol. Inst. RAS, Moscow, 1969, 400 pp.
[24] G.A. Davis, W. Cong, Y.A. Dong, J.J. Zhang, C.H. Chou, G.E. Gehrels, The enigmatic Yinshan fold-and-thrust belt of northern China: new views on intraplate contractional styles, Geology 26 (1998) 43–46.
[25] J.D.A. Zijderveld, A.C. demagnetization of rocks — analysis of results, in: D.W. Collinson, K.M. Creer, S.K. Runcorn (Eds.), Methods in Paleomagnetism, Elsevier, Amsterdam, 1967, pp. 254–286.
[26] J.L. Kirschvink, The least squares line and plane and the analysis of the paleomagnetic data, Geophys. J. R. Astron. Soc. 62 (1980) 699–718.
[27] R.A. Fisher, Dispersion on a sphere, Proc. R. Soc. London 217 (1953) 295–305.
[28] P.L. McFadden, M.W. McElhinny, The combined analysis of remagnetization circles and direct observations in paleomagnetism, Earth Planet. Sci. Lett. 87 (1988) 152–160.
[29] M.W. McElhinny, Statistical significance of the fold test in paleomagnetism, Geophys. J.R. Astron. Soc. 8 (1964) 33–40.
[30] D.W. Collinson, Methods in Rock Magnetism and Palaeomagnetism: Techniques and Instrumentation, Chapman and Hall, London, 1983, 503 pp.
[31] T. Nagata, S. Akimoto, S. Uyeda, Reverse thermo-remanent magnetism , Proc. Jpn. Acad. Sci. 27 (1951) 643–645.
[32] M. Haag, F. Heller, J.C. Carracedo, V. Soler, Remanent magnetization of andesitic and dacitic pumice from the 1985 eruption of Nevado del Ruiz (Colombia) reversed due to self-reversal, J. Volcanol. Geotherm. Res. 41 (1990) 369–377.
[33] M. Ozima, M. Funaki, N. Hamada, S. Aramaki, T. Fujii, Self-reversal of thermo-remanent magnetization in pyroclastics from the 1991 eruption of Mt. Pinatubo, Philippines, J. Geomagn. Geoelectr. 44 (1992) 979–984.
[34] P.L. McFadden, F.J. Lowes, The discrimination of mean directions drawn from Fisher distributions, Geophys. J.R. Astron. Soc. 67 (1981) 19–33.
[35] P.L. McFadden, M.W. McElhinny, A physical model for palaeosecular variation, Geophys. J.R. Astron. Soc. 78 (1984) 809–830.
[36] E. Irving, G. Pullaiah, Reversals of the geomagnetic field, magnetostratigraphy, and relative magnitude of paleosecular variation in the Phanerozoic, Earth Sci. Rev. 12 (1976) 35–64.
[37] P. Pruner, Palaeomagnetism and palaeogeography of Mongolia in the Cretaceous, Permian, and Carboniferous — preliminary data, Tectonophysics 139 (1988) 155–167.
[38] M.I. Kuzmin, V.A. Kravchinsky, First paleomagnetic data on the Okhotsk–Mongolian belt, Russ. J. Geol. Geophys. 37 (1996) 48–55.
[39] K.N. Huang, N.D. Opdyke, Paleomagnetism of Cretaceous to Lower Tertiary rocks from southern Sichuan: a revisit, Earth Planet. Sci. Lett. 112 (1992) 29–40.
[40] V. Hsu, Paleomagnetic results from one of the red basins in south China (abstract), Eos 68 (1987) 295.
[41] S.A. Gilder, R.S. Coe, H.R. Wu, G.D. Kuang, X.X. Zhao, Q. Wu, X.Z. Tang, Cretaceous and Tertiary paleomagnetic results from southeast China and their tectonic implications, Earth Planet. Sci. Lett. 117 (1993) 637–652.
[42] Y.Z. Zhai, M.K. Seguin, Y.X. Zhou, J.M. Dong, Y.X. Zheng, New paleomagnetic data from the Huanan block, China and Cretaceous tectonics in eastern China, Phys. Earth Planet. Inter. 73 (1993) 295–312.
[43] R.J. Enkin, V. Courtillot, L. Xing, Z. Zhang, Z. Zhuang, J. Zhang, The stationary Cretaceous paleomagnetic pole of Sichuan (South China Block), Tectonics 10 (1991) 547–559.
[44] J.L. Lin, The Apparent Polar Wander Paths for the North and South China Blocks, Ph.D. Thesis, University of California, Santa Barbara, CA, 1984, 248 pp.
[45] Z.W. Zhu, T. Hao, H. Zhao, Paleomagnetic study on the tectonic motion of the Pan-Xi block and adjacent area during the Yinzhi–Yanshan period, Acta Geophys. Sin. 31 (1988) 420–431.
[46] L.S. Chan, Paleomagnetism of late Mesozoic granitic intrusions in Hong Kong: implications for upper Cretaceous reference pole of south China, J. Geophys. Res. 96 (1991) 327–335.
[47] Z. Zheng, M. Kono, H. Tsunakawa, G. Kimura, Q. Wei, X. Zhu, T. Hao, The apparent polar wander path for the North China Block since the Jurassic, Geophys. J. Int. 104 (1991) 29–40.
[48] H. Wu, L. Zhou, Z. Zhao, Z. Yang, Y. Chen, Tectonic implications of the paleomagnetic results of the latter Paleozoic and Mesozoic rocks from the Alashan area of the Western North China Block, Sci. Geol. Sin. 2 (1993) 19–46.
[49] X. Ma, Z. Yang, L. Xing, The Lower Cretaceous reference pole for North China, and its tectonic implications, Geophys. J. Int. 115 (1993) 323–331.